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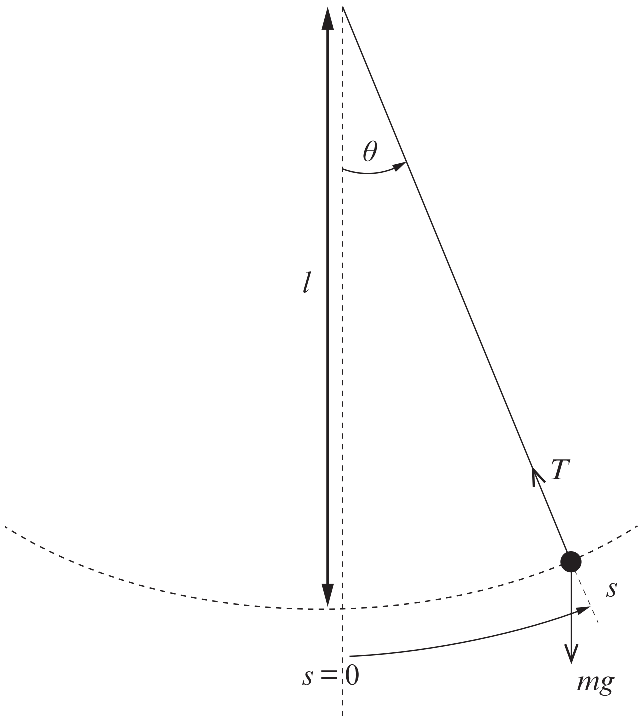
6. *(a)* The pendulum in the figure below has a small bob of mass, m, suspended at the bottom of a light string of length, l. The string is shown at an angle, θ, to the vertical and the mass, m, is at a distance, s, along the arc. The forces acting on the mass are shown.



(i) Name the **two** forces acting on the mass. [1]

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(ii) By considering these forces show that:

resultant force component on the mass along the arc = – $mg \sin\theta$

You may add to the diagram if you wish. [3]

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(iii) If the oscillation is small so that $\sin\theta \simeq \theta$ show in clear steps that the acceleration along the arc may be written as: [2]

$$\text{acceleration} = -\frac{gs}{l}$$

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(iv) Discuss whether the equation in part *(a)*(iii) satisfies the definition of simple harmonic motion. [2]

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(b) A small mass oscillates at the lower end of a pendulum of length 1.20 m.

(i) Determine its:

I. period; [2]

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II. frequency. [1]

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(ii) If the maximum displacement angle of the mass is 0.067 rad, justify the use of simple harmonic motion in part *(b)*(i). [2]

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7. *(a)* A pendulum oscillating with simple harmonic motion consists of a small 0.05 kg mass oscillating at the end of a string of length 4.0 m. The displacement x in metres at time t can be written as:

$$x = 0.25 \cos \omega t$$

(i) Show that the angular velocity, ω, is approximately 1.57 rad s^{-1}. [2]

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(ii) Calculate the maximum speed of the mass. [2]

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(iii) Show that the kinetic energy (E_k) of the mass, in joules, may be written as: [2]

$$E_k = 3.8 \times 10^{-3} \sin^2 1.57t$$

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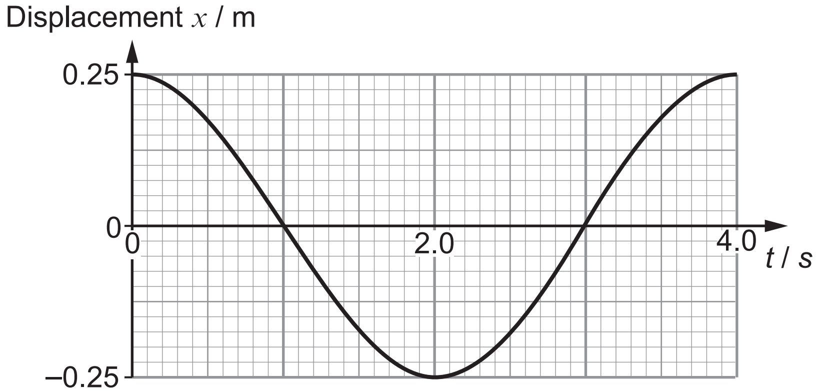
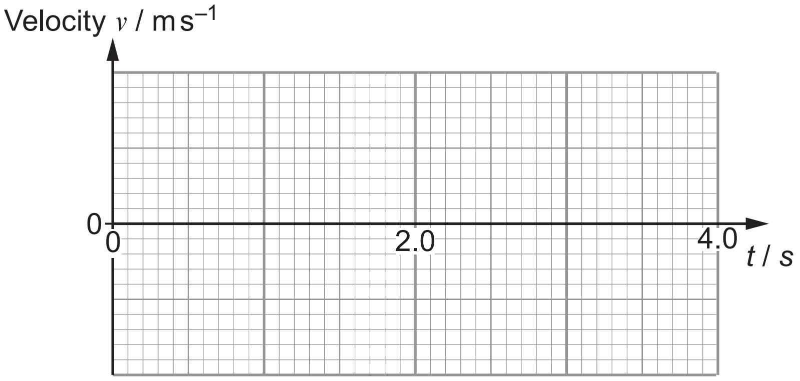
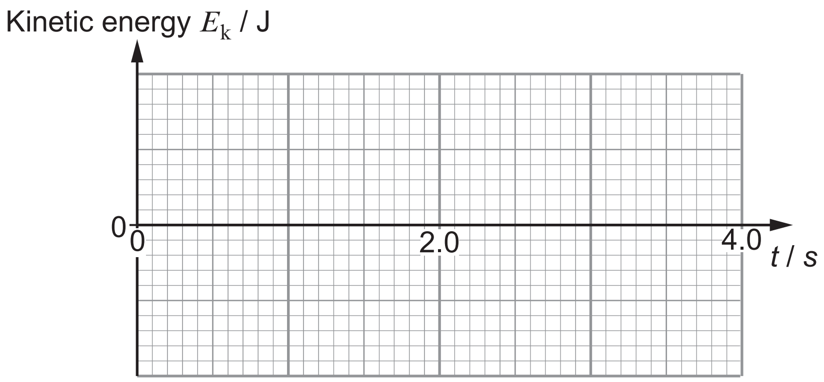
(iv) The displacement-time curve for the pendulum is shown for a time interval of one period i.e. 4.0 s. Draw curves for velocity, v, and E_k on the axes below for the same interval of time. Indicate values on your axes for v and E_k. [4]







(v) Determine the maximum height reached by the pendulum mass above the level of its lowest point. [2]

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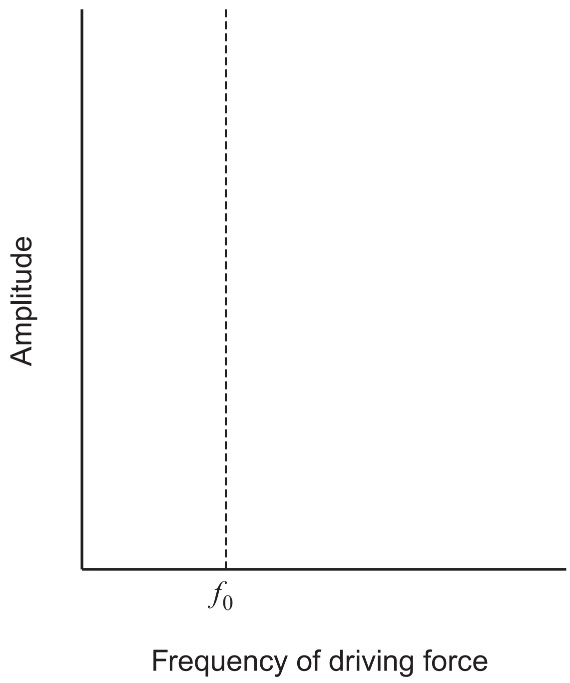
(b) A system that can oscillate may be driven by an external sinusoidal force.

(i) Sketch the amplitude of a lightly damped system as the frequency of the driver is increased. The natural frequency f_0 of the system is indicated. Label the curve **X**. [1]



(ii) Name the effect seen near the frequency, f_0. [1]

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(iii) Sketch a second curve on the **axes above** for an identical oscillating system that is damped to a greater extent. Label the curve as **Y**. [2]

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3. Simple harmonic motion (SHM) may be represented by:

$$a = -\omega^2 x$$

(a) State what quantities are represented by: [1]

a ....

ω ....

x ....

(b) If the amplitude of the oscillation of an object moving with SHM is 0.012 m and the period is 0.40 s:

(i) determine the maximum acceleration; [3]

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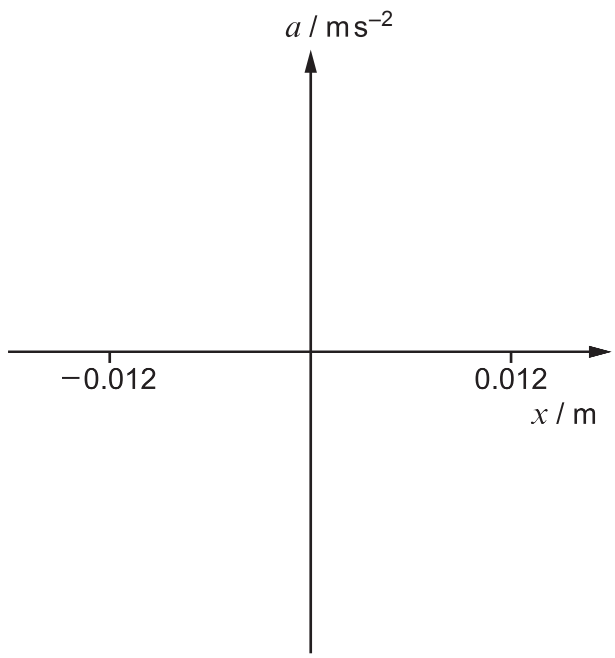
(ii) sketch a graph showing the variation of a with x. Insert values on the a-axis. Values are given on the x-axis. [3]



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(c) The oscillating object in part *(b)* is at its centre position when t = 0 and moving in the positive x direction. Insert the missing values in the expression for x in metres. [3]

$$x = \boxed{}\cos(\boxed{}\,t + \boxed{})$$

(d) An oscillating system may be driven by an external force. Describe and explain:

(i) an application of forced oscillations that is useful; [2]

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(ii) an example of forced oscillations that should be avoided. [3]

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4. Describe an experiment that uses a pendulum to determine the acceleration due to gravity with graphical analysis. [6 QER]

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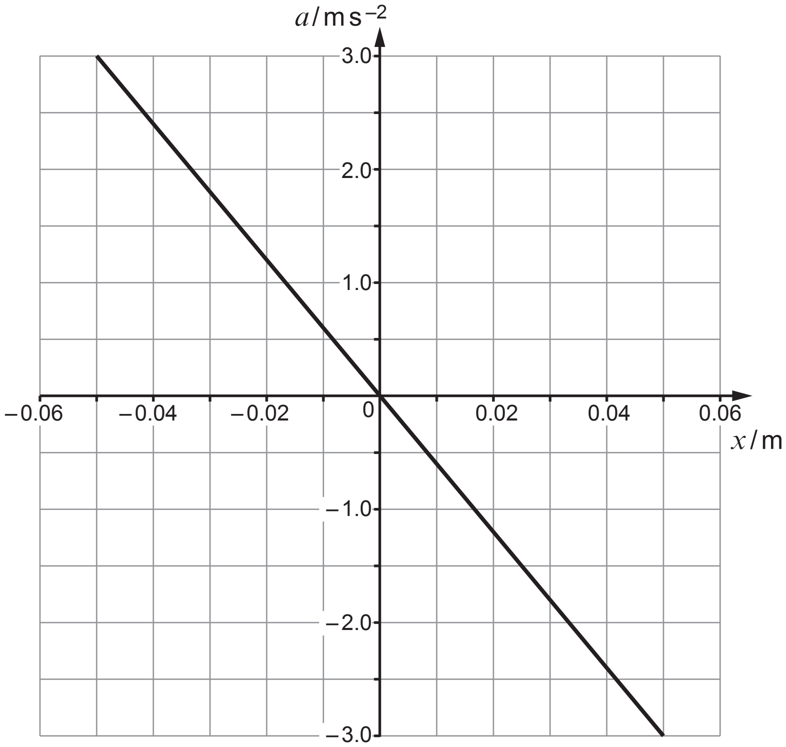
3. A small object of mass 0.30 kg oscillates at the end of a spring. The graph shows how its acceleration, a, depends on its displacement, x, from a fixed point.



(a) (i) Identify the **two** characteristics **of the graph** that show that the object is oscillating with *simple harmonic motion.* [2]

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(ii) Use the graph to show that the angular frequency, ω, of the oscillation is approximately 7.7 rad s^{-1}. [3]

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(iii) Calculate:

I. the period of oscillation; [1]

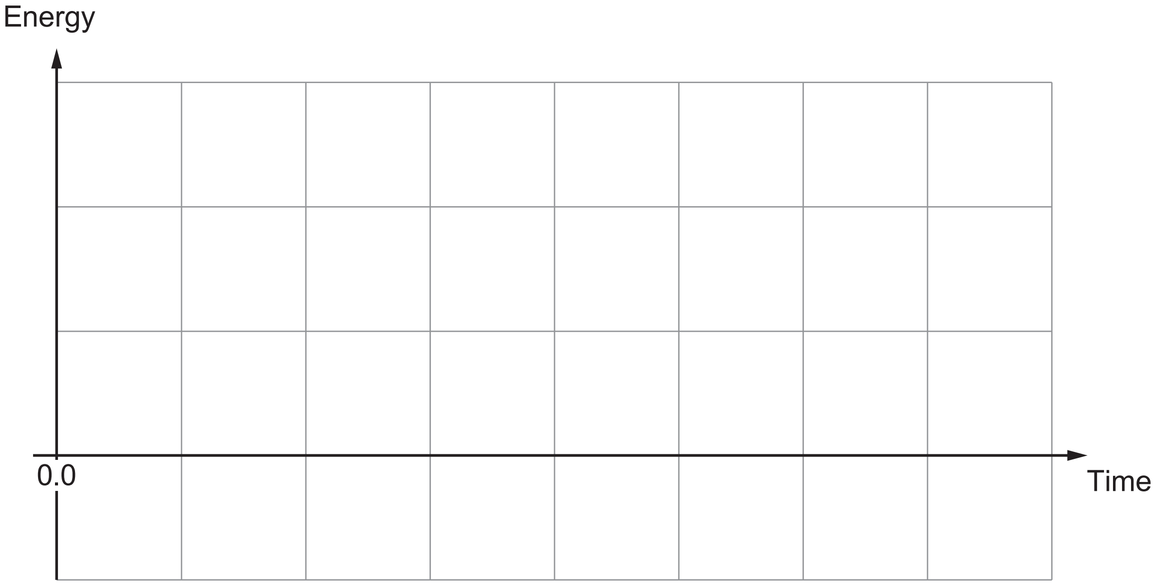
II. the spring constant; [2]

III. the velocity of the object as it passes through the centre of oscillation. [2]

(iv) The object was released at the displacement of –0.05 m at time t = 0. Sketch on the axes below:

I. the potential energy of the system (label the sketch PE),
II. the kinetic energy of the system (label the sketch KE).

Draw both sketches **over one cycle** of the undamped oscillation. (No values are required on the axes.) [3]



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(b) Explain what is meant by the terms *damping* and *resonance* **and** discuss their importance in real systems giving an example for each. [6 QER]

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6. A student says that a simple pendulum oscillates with simple harmonic motion with a period, T, of:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of the pendulum and g is the acceleration due to gravity.

(a) Describe what is meant by a simple pendulum. [1]

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(b) State what is meant by simple harmonic motion. [2]

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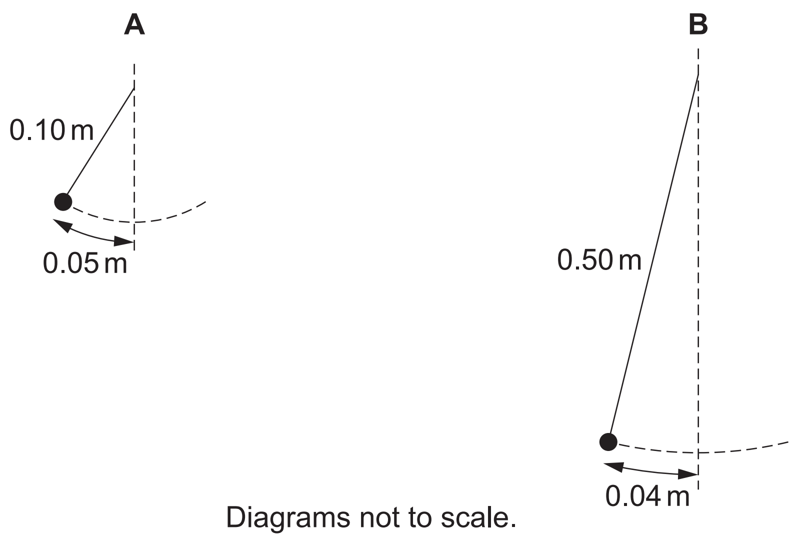
(c) The student sets up two experimental systems to investigate simple harmonic motion. System A is a pendulum of length 0.10 m oscillating with an amplitude of 0.05 m. System B is a pendulum of length 0.50 m oscillating with an amplitude of 0.04 m. Give two reasons why system B is better than system A to investigate simple harmonic motion. [2]



Diagrams not to scale.

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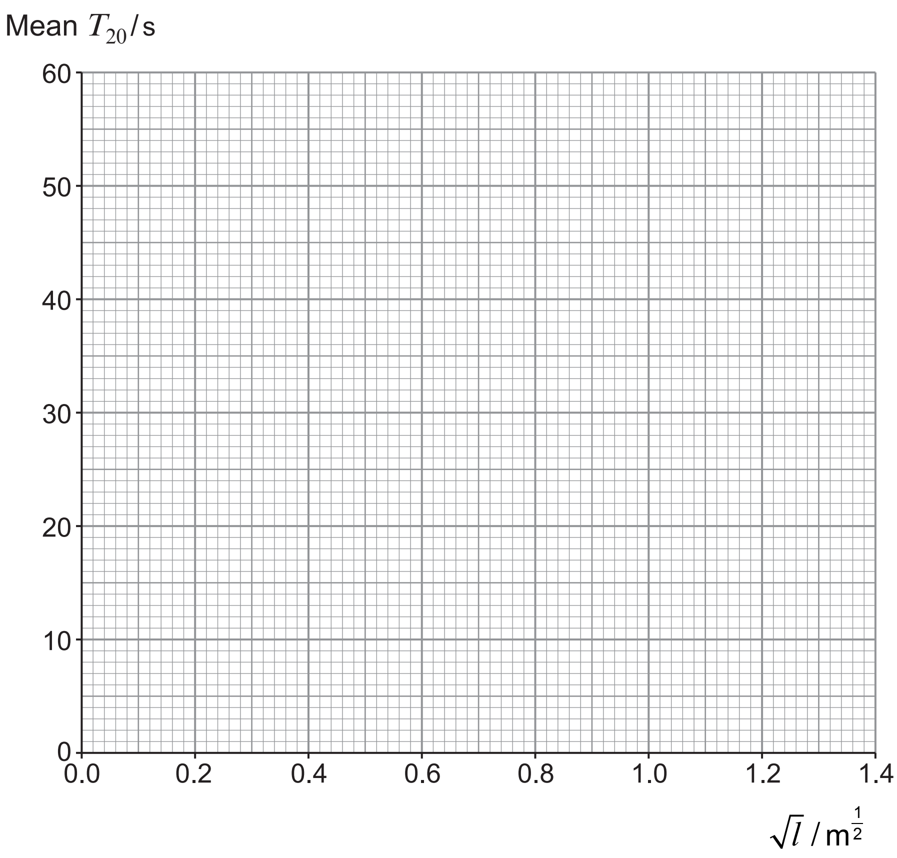
Examiner only

(d) Measurements for an oscillating simple pendulum are given in the table below, where l is the length of the pendulum and T_{20} is the time taken for 20 oscillations. For each value of l four values of T_{20} are measured.

l / m	$\sqrt{l}$ / $m^{\frac{1}{2}}$	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s
0.250	0.50	19, 21, 22, 20	21	2
0.500	0.71	26, 29, 26, 29	28	2
0.750	....	35, 34, 33, 36	....	....
1.000	....	42, 39, 40, 41	....	....
1.250	....	45, 42, 44, 45	....	....
1.500	....	48, 50, 49, 52	....	....

(i) **Complete the table** by determining $\sqrt{l}$, the mean time for 20 oscillations, T_{20} and the uncertainty in T_{20}. Values have been inserted for the first two lengths. [3]

(ii) Plot mean T_{20} against $\sqrt{l}$ on the grid below. Include error bars on the T_{20} axis only. Draw a line of maximum gradient and a line of minimum gradient. [4]



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(iii) Justify why error bars for $\sqrt{l}$ are not plotted on the graph. [2]

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(iv) Use the maximum and minimum gradients to determine a mean value for the acceleration due to gravity, g, and its **percentage** uncertainty. [5]

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(e) Another student says that the gradient of the graph would be approximately twice as large if the experiment was carried out on the Moon, where the acceleration is approximately $0.2g$. Investigate if this statement is correct, justifying your answer. [3]

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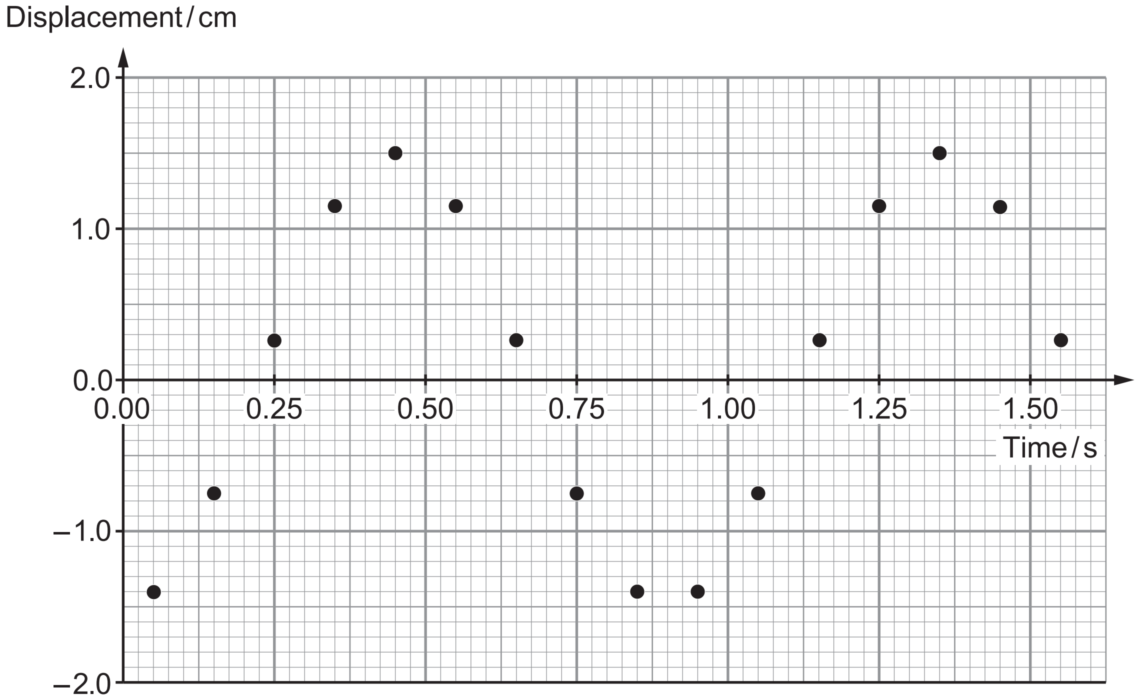
4. A 0.030 kg mass is suspended from a light, vertical spring. A student sets the mass to oscillate vertically. The position of the oscillating mass is tracked using a datalogger. Measurements, taken over a time interval of approximately 1.50 s, are shown on the graph below. The amplitude during this interval was 1.5 cm.



(a) **Draw on the grid** above the curve of best fit. [2]

(b) Explain what is meant by the 'period of the oscillation'. Use the graph to support your answer. [2]

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(c) **Complete** the expression for the displacement, y, of the mass. [2]

$$y = 1.5\cos\left(\left(\frac{2\pi}{\dots\dots}\right)t \quad \dots\dots\dots\dots\right)\ \text{cm}$$

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(d) (i) By drawing a tangent to the displacement-time curve, show that the maximum velocity of the oscillating mass is approximately 10 cm s^{-1}. [3]

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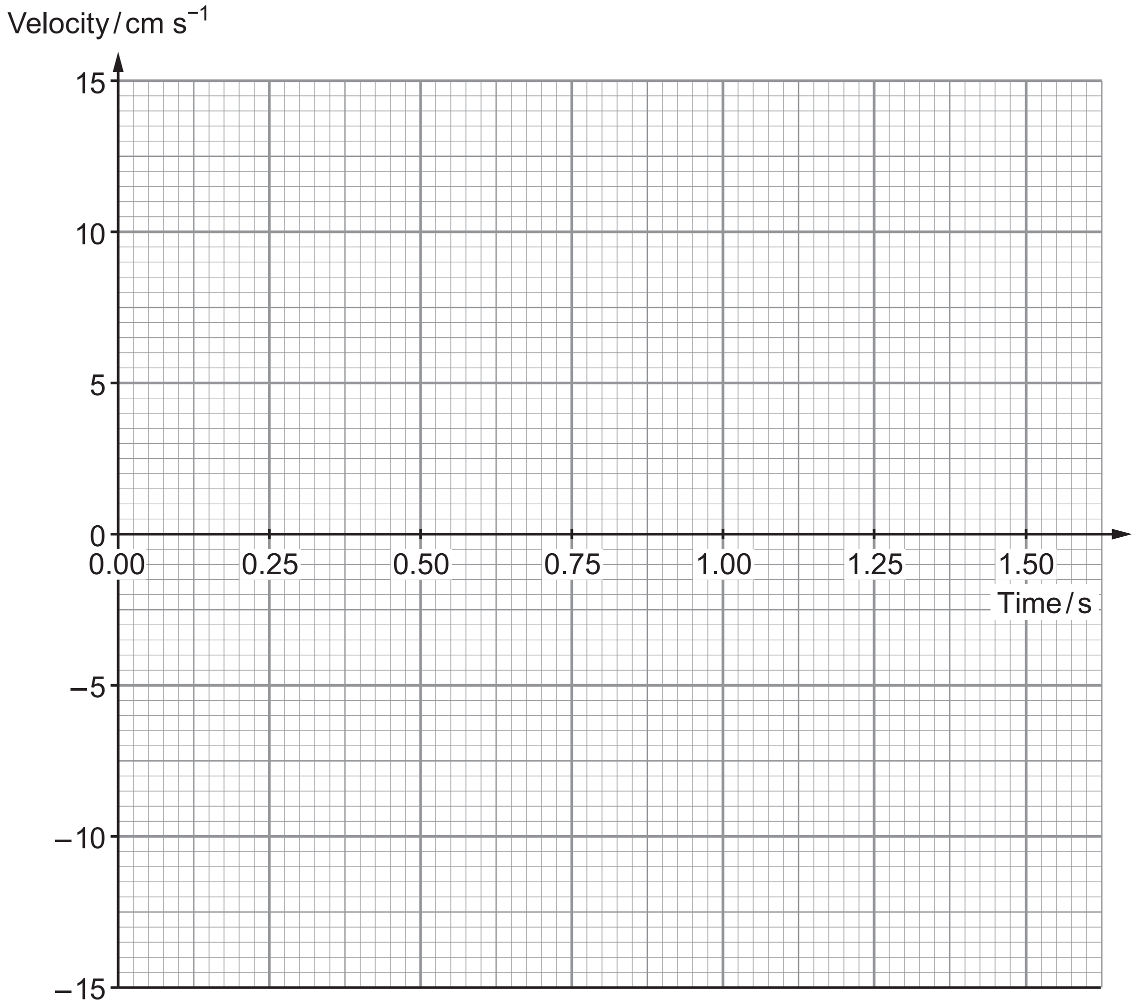
(ii) **Sketch** the velocity-time graph for the oscillating mass on the grid below. [3]



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(e) Catrin suggests that the 0.030 kg mass is oscillating with simple harmonic motion.

(i) State what is meant by 'simple harmonic motion'. [2]

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(ii) Calculate the value of the spring constant. [3]

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Examiner only

(f) Catrin also claims that the maximum elastic potential energy stored by the spring is far greater than the maximum kinetic energy of the mass. Evaluate whether or not this claim is correct.

Amplitude of the motion = 1.5 cm

Mass = 0.030 kg

Maximum speed = approximately 10 cm s^{-1} [6]

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5. Bethan investigates damped oscillations and notices that the amplitude of an oscillating system decreases from 1.5 ± 0.1 cm to 0.5 ± 0.1 cm in 12 minutes.

(a) Suggest what causes the decrease in amplitude. [1]

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(b) This reduction in amplitude may be represented by:

$$A = A_0 e^{-\lambda t}$$

where A is the amplitude at time t, A_0 is the initial amplitude, and λ is a characteristic constant for the system.
By determining the maximum and minimum possible values of λ, determine its value and its **absolute** uncertainty. [5]

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(c) Give **one** example of damped motion that occurs in everyday life. [1]

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